

Anscombe's Quartet

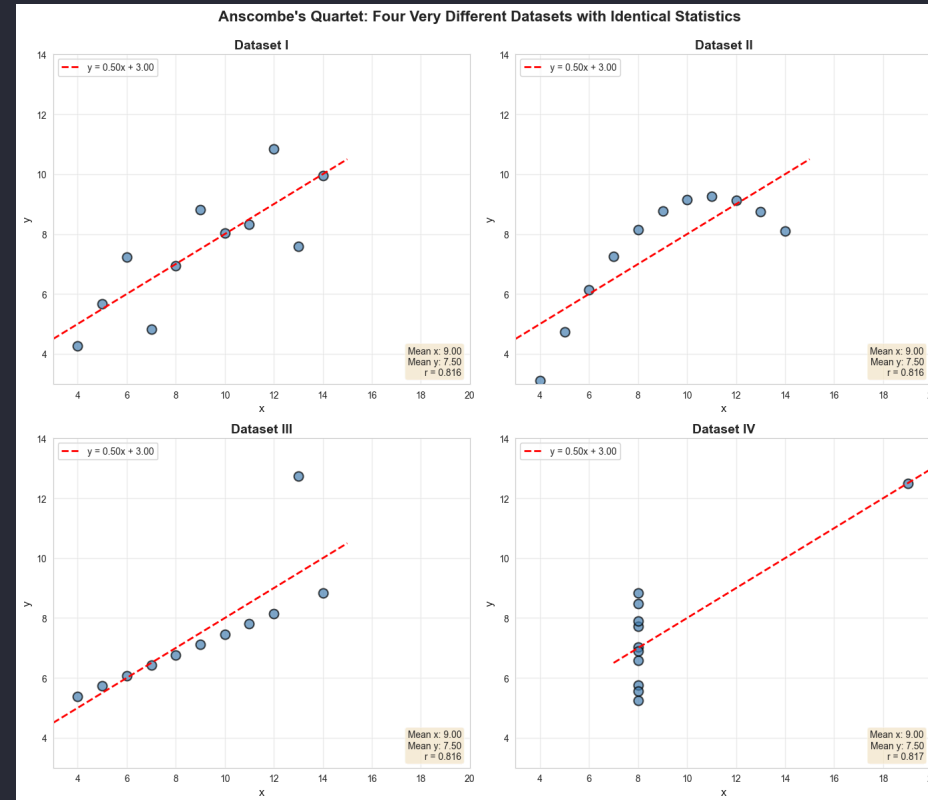


The Setup: Four "Identical" Datasets

- All four datasets have **exactly the same** statistics:
 - Mean of X: **9.0**
 - Mean of Y: **7.5**
 - Standard deviation of X: **3.3**
 - Standard deviation of Y: **2.0**
 - Correlation: **0.816**
 - Linear regression: **$y = 0.5x + 3.0$**

“ **Question:** *If the statistics are identical, are the datasets the same?* ”

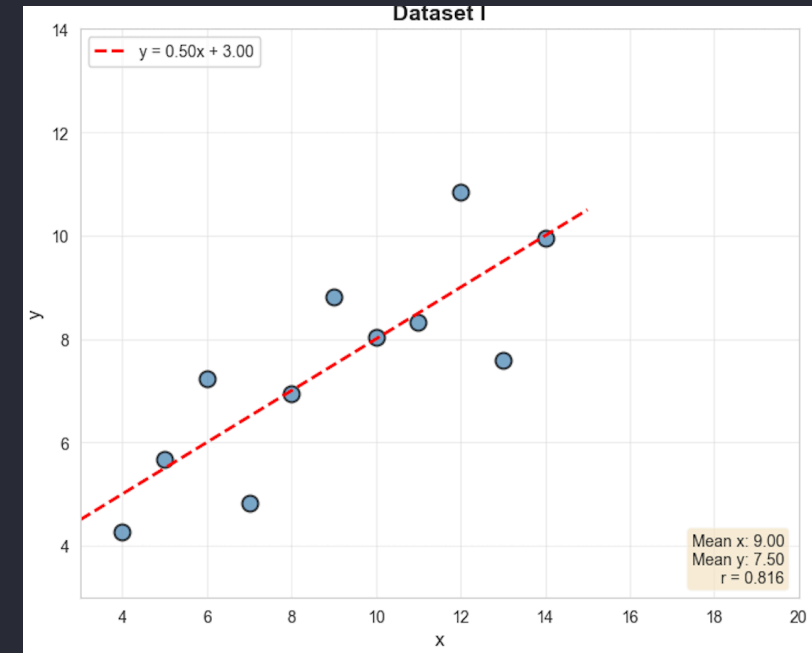
The Reality: Four Completely Different Stories



Always visualize your data!

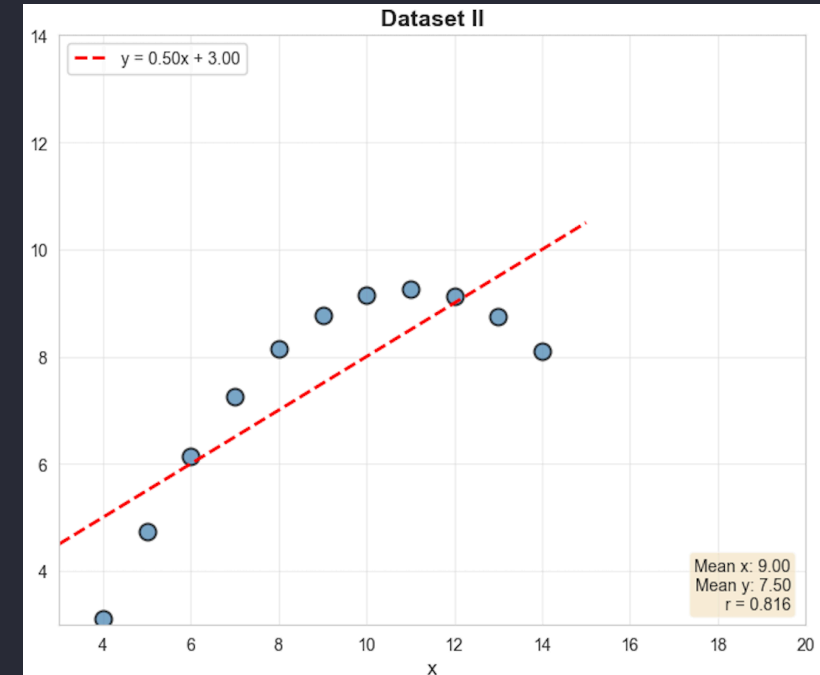
Dataset I: The "Good" One

- **Pattern:** Simple linear relationship with random scatter
- **What it means:**
 - Linear regression is appropriate
 - Predictions will be reasonably accurate
 - This is what we *hope* our data looks like
- **Takeaway:** Even perfect scenarios have variance!



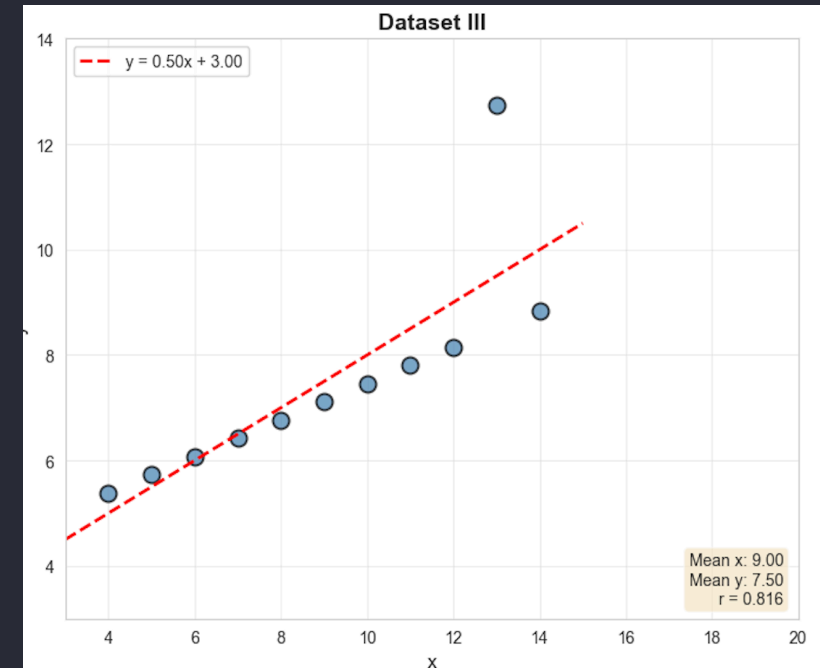
Dataset II: The Curved One

- **Pattern:** Clear non-linear (quadratic) relationship
- **The Problem:**
 - Linear model makes **systematic errors**
 - Underestimates at edges, overestimates in middle
 - U-shaped residual pattern (red flag!)
- **The Fix:**
 - Use polynomial or non-linear regression
 - $R^2 = 0.67$ looks "good" but model is **wrong**



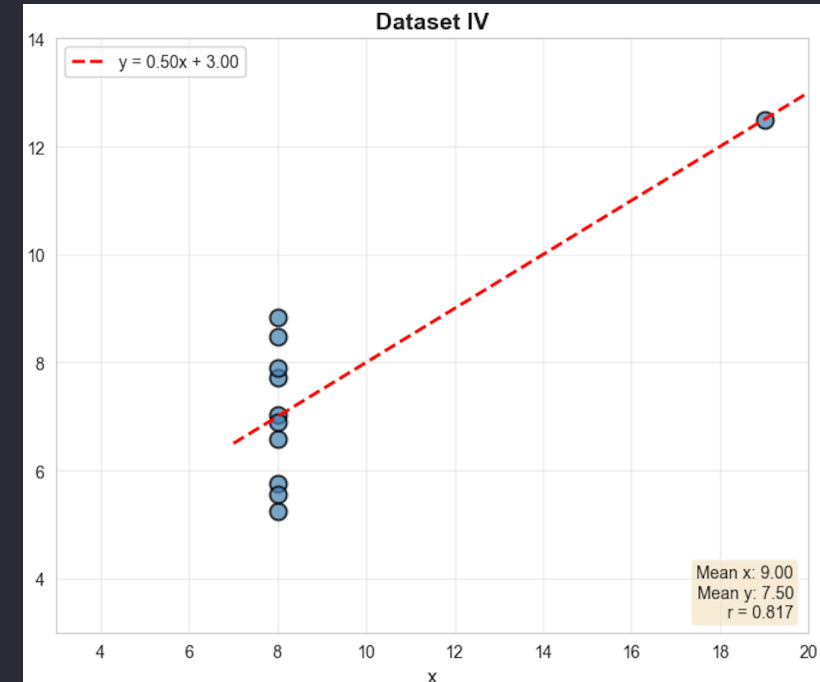
Dataset III: The Outlier

- **Pattern:** Perfect linear... except ONE point
- **The Problem:**
 - Single outlier drastically affects the regression line
 - Rest of data follows a different pattern
 - Outlier has high **leverage** (unusual x-value)
- **Questions to Ask:**
 - Is this a data error?
 - Is this a special case to investigate?
 - Should we use robust regression?



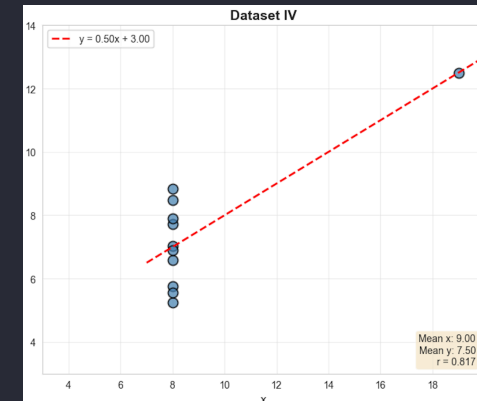
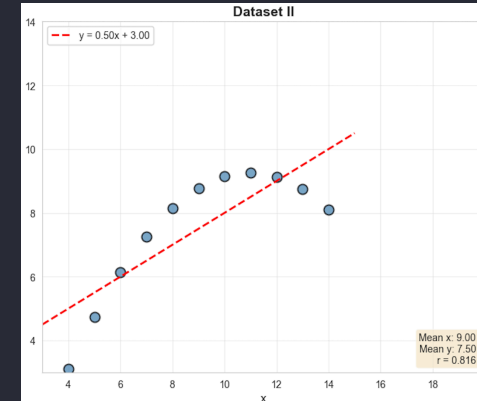
Dataset IV: The Most Dangerous

- **Pattern:** NO relationship exists!
- **The Illusion:**
 - 10/11 points have $x = 8$ (no variance!)
 - ONE outlier at $x = 19$ creates fake correlation
 - Remove it → correlation drops to ~ 0
- **Real-World Impact:**
 - Would conclude "X predicts Y" when it doesn't
 - Business decisions based on false relationships
 - Potentially catastrophic outcomes



The Business Scenario

- **Imagine:** Y = Sales, X = Ad Spend
- **Dataset II** (non-linear):
 - Wrong model $\rightarrow$ wrong budget allocation
 - Miss optimal spending levels
- **Dataset IV** (outlier-driven):
 - Believe ads drive sales when they don't!
 - Waste millions on ineffective campaigns
 - Miss actual drivers of sales
- **Cost:** Potentially millions in poor decisions



Lesson 1: Summary Statistics Lie

- **Same numbers can tell completely different stories**
 - Mean, std dev, correlation can be identical
 - Yet underlying data is fundamentally different
 - Numbers without context are dangerous
- **Golden Rule:** Never trust statistics alone

Lesson 2: Visualization is Non-Negotiable

- **Visualization reveals what statistics hide:**
 - Non-linear relationships
 - Outliers and anomalies
 - Clusters and groupings
 - Data quality issues
 - Violations of assumptions
- **Best Practice:** Plot first, model second

Lesson 3: Check Your Assumptions

- Linear regression assumes:
 1. **Linearity** - relationship is actually linear
 2. **Independence** - observations are independent
 3. **Homoscedasticity** - constant variance
 4. **Normality** - residuals are normally distributed
 5. **No outliers** - or at least account for them
- **If assumptions fail, your predictions fail**

Lesson 4: Residual Plots Are Your Friend

- **Good residual plot:**
 - Random scatter around zero
 - No patterns or trends
 - Constant spread
- **Bad residual plots reveal:**
 - U-shape → non-linear relationship (Dataset II)
 - Funnel shape → heteroscedasticity
 - Clusters → missing variables
 - Outliers → special cases

Lesson 5: One Outlier Can Ruin Everything

- **High leverage points** (unusual x-values) have outsized influence:
 - Can create relationships that don't exist
 - Can hide relationships that do exist
 - Can dramatically change model parameters
- **What to do:**
 1. **Investigate** - Why is this point different?
 2. **Don't auto-delete** - It might be your most important finding!
 3. **Consider robust methods** - Less sensitive to outliers
 4. **Report sensitivity** - How much does model change?

Lesson 6: Good Metrics $\neq$ Good Model

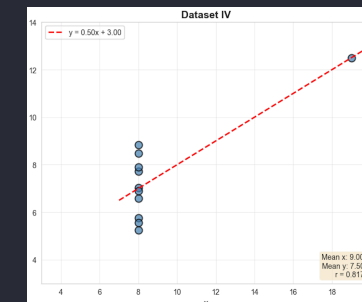
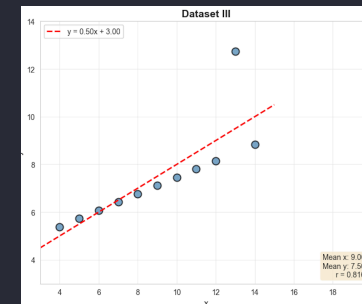
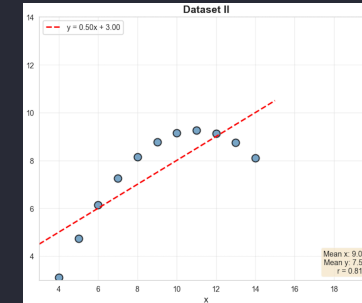
- **Dataset II:** $R^2 = 0.67$ (looks decent!)
 - But model is **fundamentally wrong**
 - Systematic prediction errors
 - Choosing wrong model type
- **Dataset IV:** $R^2 = 0.67$ (looks decent!)
 - But relationship is **completely false**
 - Driven by single outlier
 - Would make terrible predictions
- **Beware:** Metrics can be misleading without context

Best Practices

1. Visualize First
2. Explore & Understand
3. State Assumptions
4. Build Model
5. Diagnostic Checks (residuals, outliers)
6. Validate on New Data
7. Maintain Skepticism!

Real-World Applications

- **Dataset II (Non-linear):**
 - Returns diminishing over scale
 - Learning curves, dose-response
 - Economic relationships
- **Dataset III (Outlier):**
 - Rare events or anomalies
 - Data entry errors
 - Special cases worth studying
- **Dataset IV (False correlation):**
 - Confounding variables
 - Spurious relationships
 - Small sample sizes



Common Mistakes to Avoid

- Reporting only R^2 without visualizing
- Deleting outliers without investigation
- Assuming linearity without checking
- Ignoring residual patterns
- Using models on data they weren't built for
- Trusting a model just because it has "good" metrics
- Forgetting to validate on new data

What Makes a Good Data Scientist?

- **Technical Skills:**

- Know your statistical methods
- Understand assumptions and limitations
- Use diagnostic tools effectively

- **Critical Thinking:**

- Question the data and results
- Look for what might be wrong
- Don't fall in love with your models

- **Communication:**

- Visualize to explain
- Acknowledge uncertainty
- Document assumptions

The Anscombe's Quartet Mindset

“ *"The greatest value of a picture is when it forces us to notice what we never expected to see."*

— *John Tukey*

”

“ *Every dataset is guilty until proven innocent by visualization*

”

Key Takeaway

- In data science, seeing is believing
- Summary statistics are never enough
- Always **Visualize** your data

Video

- Same Stats, Different Graphs

References

Topic	Source
Anscombe (1973)	Anscombe, F. J. (1973). Graphs in Statistical Analysis . <i>The American Statistician</i> , 27(1), 17–21.
Tukey (1977)	Tukey, J. W. (1977). <i>Exploratory Data Analysis</i> . Addison-Wesley.